The bridge between Cody and Codex

CCP discovers a device, saves a trusted pairing, and checks that its paired PC is still there.

Cody's Codex Plugin

A small Python service on your Windows PC. Codex calls its local tools; Cody talks to it over the LAN.

Cody finds CCP, not arbitrary Codex windows or chats.



01 / Discover

Find pairing-ready Cody devices without typing an IP address.

02 / Pair

Use the display code to retain the intended PC/device relationship.

03 / Stay in touch

Authenticate repeated replies from the retained PC identity.

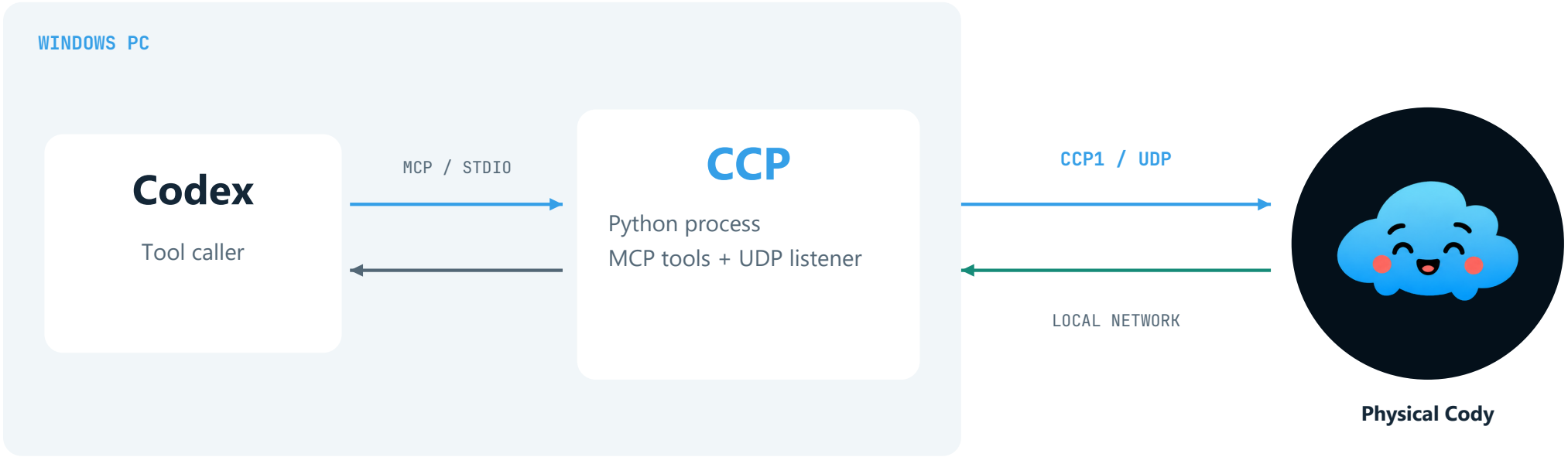
SOURCE / CCP1.md; README.md

NOTES / QUESTIONS

Copyright 2026 / confidential

Two links. One local bridge.

The Codex-to-plugin connection is separate from the plugin-to-device network protocol.



Tools exposed to Codex

ccp_discover_devices
ccp_pair_device
ccp_connection_status

Not implemented by this link

No Codex chat selection, task stream or project/message transport. No raw Codex service is exposed on the LAN.

SOURCE / mcp.json; ccp_server.py: CcpService, CompanionResponder

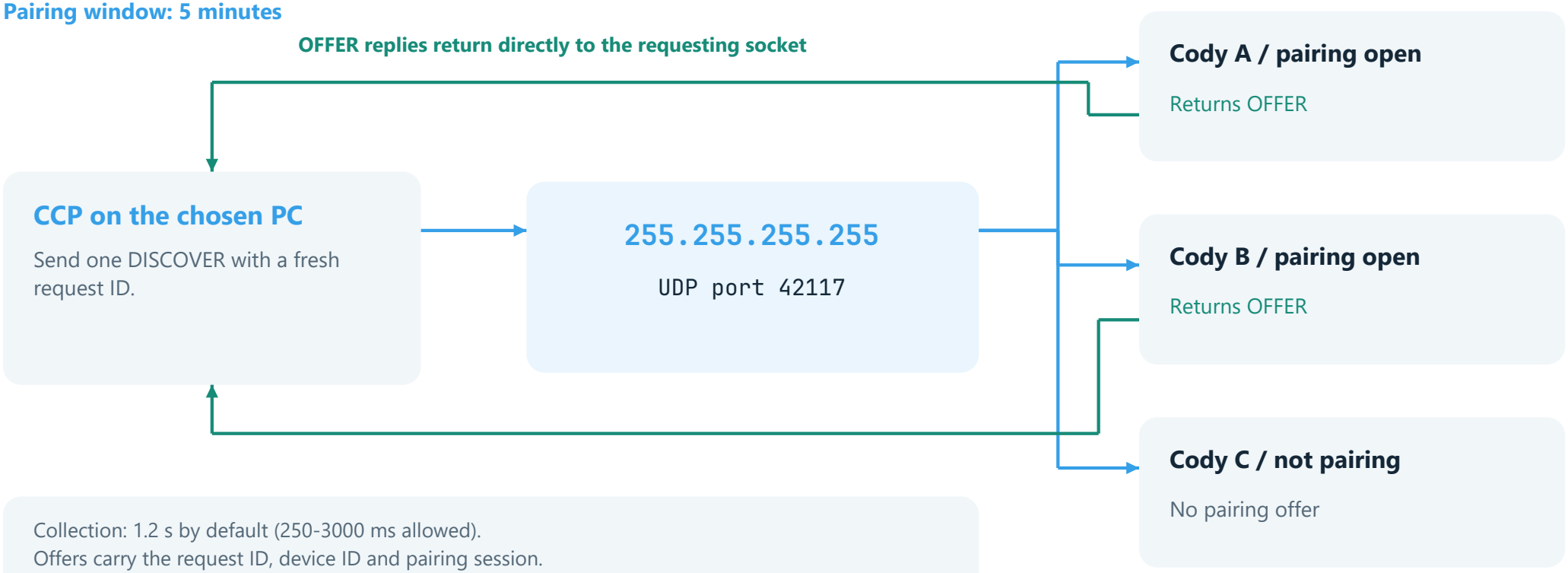
NOTES / QUESTIONS

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First discovery: the PC asks the LAN

Only devices with an open pairing window offer themselves for pairing.

Pairing window: 5 minutes



Collection: 1.2 s by default (250-3000 ms allowed).
Offers carry the request ID, device ID and pairing session.
The IP address is observed from the reply, not entered by you.

SOURCE / ccp_server.py: discover; LUNASHELL_codex_link.cpp: sendOffer

NOTES / QUESTIONS

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The six characters select the right Cody

An offer is a candidate, not permission to pair. CCP must verify it using the user's code.

DISPLAY CODE

K7M4Q2

→ **Code + each offer's salt**

PBKDF2-HMAC-SHA256 / 50,000 iterations

EXAMPLE CODE

Offer A

Recompute its HMAC tag using the derived verifier.

NO MATCH

Offer B

Recompute its HMAC tag using the derived verifier.

MATCH

Offer C

Recompute its HMAC tag using the derived verifier.

NO MATCH

The code is not sent in clear text in CCP1 LAN datagrams.

No match

Stop. Check the code or refresh the offer.

Exactly one match

Continue with this device and session.

More than one match

Stop. Restart pairing; never pick the first.

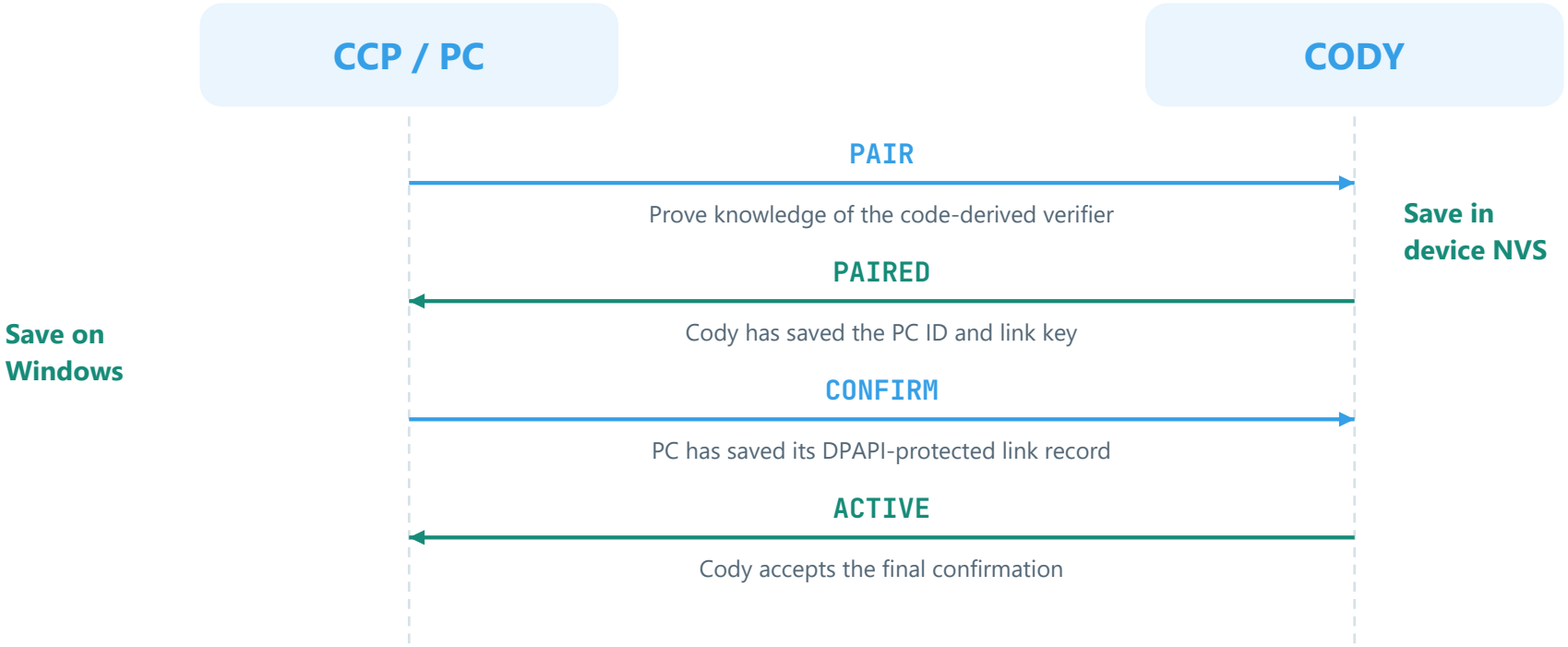
SOURCE / CCP1.md: Pairing; ccp_server.py: _matching_offers, pair

NOTES / QUESTIONS

Copyright 2026 / confidential

Pairing is complete only after confirmation

Cody and the PC each persist the relationship before the plugin reports successful pairing.



PAIR and CONFIRM retry every 0.5 s within separate 3 s windows. A short ACTIVE replay window helps recover a lost final reply. A timeout is not reported as success.

SOURCE / CCP1.md: PAIR, PAIRED, CONFIRM, ACTIVE

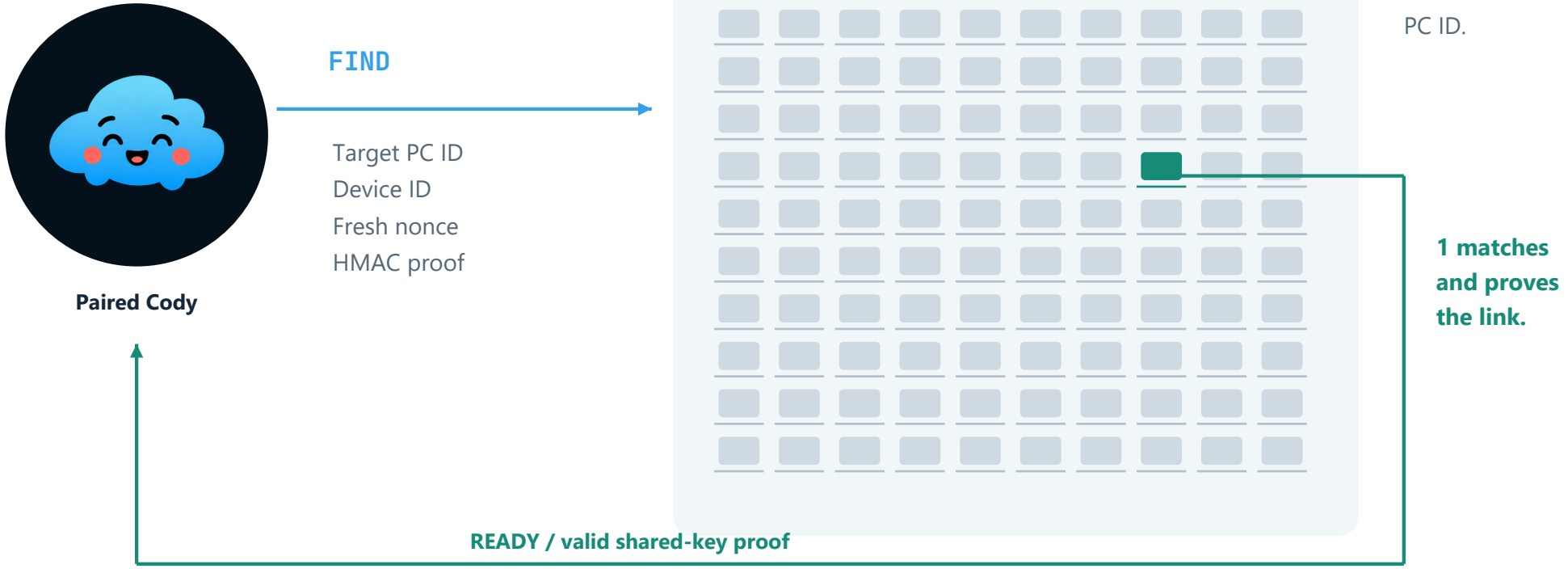
NOTES / QUESTIONS

Copyright 2026 / confidential

A hundred PCs. One retained identity.

Broadcast delivery reaches the local network; identity and proof decide who may answer.

IP addresses may change;
the paired identity stays.



Illustrative 100-PC network / not a load-test result

SOURCE / ccp_server.py: CompanionResponder._handle_find; StateStore

NOTES / QUESTIONS

Copyright 2026 / confidential

Rediscovery starts from Cody

After setup, the saved PC identity replaces the temporary display code.

13
Confirmed
Tap Let's go

→

16
Idle
10 active seconds

→

14
Find paired PC
FIND / READY

→

15
Connected
CCP is ready

Cody sends FIND
Broadcast to UDP 42118. Include the retained PC ID, device ID, fresh nonce and shared-key proof.

→

The paired CCP returns READY
Reply directly to Cody. Echo the nonce and authenticate the reply with the same retained link key.

←

Cody accepts only the expected device ID, PC ID, current nonce and proof. It learns the PC's current address from the verified reply.

SOURCE / LUNASHELL_codex_link.cpp: sendFind, acceptReady; Cody onboarding flow

NOTES / QUESTIONS

Copyright 2026 / confidential

Connected is a renewable observation

The device keeps checking; a saved pairing alone is not evidence that the PC is online.

HEALTHY LINK



NO MORE VALID REPLIES



Device view

FIND every 2 s. After 8 s without a valid READY, Cody returns to discovery.

PC status view

CCP uses a separate 10 s last-seen threshold. This is not a task/session health check.

SOURCE / LUNASHELL_codex_link.cpp: maintainState; ccp_server.py: public_status

NOTES / QUESTIONS

Copyright 2026 / confidential

Trust has a clearly defined boundary

Code-derived authentication protects selection; this is not a production security certification.

Implemented protections

A fresh pairing salt and session.
HMAC proofs bind identities and nonces.
Exactly one code-matching offer is required.
PC keys are protected by Windows DPAPI.
Cody retains the link in a versioned NVS record.

Important qualifications

The protocol is not a PAKE or TLS channel.
Captured code-derived proofs permit offline guesses; PBKDF2 increases their cost.
The NVS CRC detects record corruption; it is not encryption of the stored key.

What travels where

You enter the code into Codex, which passes it to CCP. CCP1 LAN packets contain identifiers and proofs, not the clear-text code. This is a LAN-protocol statement, not a claim about chat history or Codex data handling.

Production review still needs to cover code-guessing resistance, replay/rate limits, device key storage and hostile-network behavior.

SOURCE / CCP1.md: Security boundary; ccp_server.py: derive_verifier, StateStore

NOTES / QUESTIONS

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What must be true on a real network

The implementation is a starting point for commissioning and liveness, not an end-to-end release claim.

Network

Same reachable IPv4 broadcast domain. Private-network UDP 42117/42118 must be allowed. Guest isolation, VLAN boundaries and VPN routing can prevent discovery.

PC runtime

Windows + Python + running CCP. One CCP process owns the fixed listener per PC. A second process fails instead of sharing that socket. Not one identity per chat.

Current scope

Discover, pair and inspect liveness. No cloud discovery directory. No per-chat selection or Codex task/message transport. Multi-adapter routing needs real-network checks.

READ FROM SOURCE

Protocol messages, ports, identity checks, persistence order and configured timers.

STILL TO VERIFY END TO END

Plugin installation and lifetime in Codex; firewall and multi-adapter behavior; real Cody pairing/reconnect; packet loss, many-device load and security review. No runtime or physical tests were performed for this document.

SOURCE / README.md; CCP1.md; ccp_server.py; LUNASHELL_codex_link.cpp

NOTES / QUESTIONS

Copyright 2026 / confidential

Startup: two saved decisions

Local Wi-Fi/IP first. Then the retained Codex pairing. Public Internet is not required.

1 / No saved Wi-Fi profile

A valid absent-profile observation opens 04 > 05 > 06 > 07 > 09. Debug may show 39 before 09. Continue: 10 > 36 > 11 (skin), then the Codex decision.

2 / Saved Wi-Fi; waiting for the network

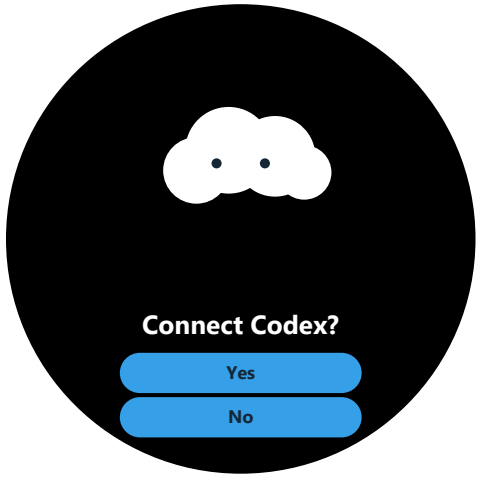
42 / No connection keeps the profile while the Wi-Fi owner reconnects. Network setup reopens 04-09; Later returns to Idle. Invalid/stale data shows UNAVAILABLE.

3 / LAN ready; no saved Codex pairing

40 / Connect Codex? Yes starts the code on 12. No returns to 16 / Idle without a settings write or CCP request. The next boot may ask again.

4 / LAN ready; saved Codex pairing

Adopt the saved attempt and search on 14. Only authenticated READY enters 15. A retained identity alone never means the PC is connected.



40 / Explicit Yes or No

Illustrative UI. Choice controls do not start pairing until the user acts.

Reopen setup from Cody

Hold the visible character still for 1.5 seconds. No button-based factory reset is added.

41 / Setup

Connect Codex opens a fresh display-code attempt. Network setup reopens 04-09. Left Back returns to Idle. Opening or leaving the menu erases no Wi-Fi, skin or pairing.

Keep the gesture deliberate

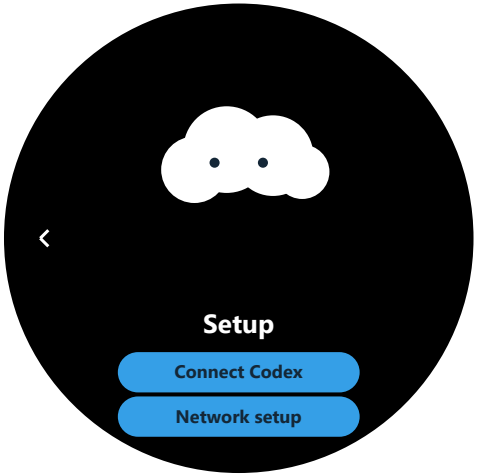
Available from stable Idle, PC search/connected and terminal CCP errors. Moving more than 12 px, a sample gap over 250 ms, rotation or a lifecycle interruption cancels the hold. Busy work and active provisioning are excluded; release before choosing.

PWR / existing policy

Single: standby/wake; active Debug defaults to screenshot countdown. Long release: standby. Debug double within 500 ms, or a continuous 3-second hold: restart preserving data.

BOOT / existing policy

Short: restore a dim/sleeping panel first, otherwise toggle fullscreen/status arcs. Hold 3 seconds: Debug restart; Release has no firmware reset action. Saved Wi-Fi and CCP survive restart.



41 / Reversible setup access

Illustrative UI. A true factory erase is a separate explicitly requested operation.

Errata and corrections

Revision date: 09 October 2026 / CODY-SETUP-20261009

Original analysis gap

The initial analysis omitted startup with Wi-Fi already configured but no saved Codex pairing, and did not define a reversible user entry to setup. This was an original analysis gap (buco di analisi iniziale), now addressed in source.

Affected original pages and screen IDs

Original pages 01 (scope), 06 (retained link) and 07 (rediscovery); screens 12-16. Existing protocol/security qualifications remain in force.

Corrections

Add 40 / optional Yes-No Codex choice, 41 / character-hold Setup, and 42 / reconnect wait. 09 means local Wi-Fi/IP; 11 selects a skin. No/Later performs no persistence or new CCP request. Saved pairing returns to 14 and requires authenticated READY for 15. PWR/BOOT do not clear pairing.

Preservation

All 10 previous pages, earlier corrections, bookmarks and unrelated content are retained. This dated addendum supersedes only the cited startup/setup/install statements; the original numbering stays stable.

Evidence and limits

Firmware source implementation and static integration review completed. PDF pages rendered and visually reviewed. No firmware tests, validation scans, configure/build/size, simulator execution, device pairing/input, flash or erase accompanied this revision. Runtime validation is pending.

Setup after standby

Hold Cody for 1.5 seconds to choose Network setup or Connect Codex.

04-06 / Escape the network loop

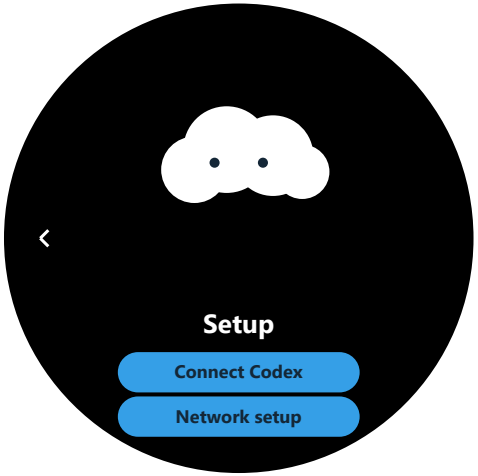
After inactivity, Welcome previously sent its first touch straight back to Let's pair. Welcome now opens 41 on a stationary tap release. A character hold also opens 41 from timeout Welcome, Let's pair or the installation QR page when no operation is busy.

35B / Wake, then hold

A finger already down during wake starts its 1.5-second hold only on a fresh acquired sample with the restored visible character. Hidden standby/wake time never counts. Movement over 12 px, report gaps over 250 ms or rotation cancel the hold. Release before choosing.

41 / Two explicit destinations

Upper Connect Codex opens 12 and its six-character CCP code using the local Wi-Fi/IP connection. Lower Network setup explicitly opens 04-09. No Internet probe is required. Active provisioning, worker admission and pending cancellation block menu access.



41 / Same choices after wake

Illustrative UI. Existing credentials and retained CCP pairing are preserved.

Errata and corrections

Revision date: 09 October 2026 / recovery correction

Original analysis gap

The first analysis did not reconcile the legacy Welcome timeout/touch path with reopening setup after standby. This was an original analysis gap (buco di analisi iniziale).

Affected pages and screens

Previous setup addendum: page 12. Affected screen IDs: 04, 05, 06, 12, 35B and 41. Earlier dated errata remain intact.

Corrections

Timeout Welcome tap release and stationary character holds on resting 04-06 reopen 41. A hold spanning wake starts only after visible restoration. Connect Codex goes to 12; Network setup goes to 04-09. Busy work and pending cleanup cannot be bypassed.

Evidence and limits

A user-authorized 90-second passive serial capture confirmed LAN/DHCP, Welcome inactivity return and standby. No Connect Codex press occurred in that capture; the initial wrong-selection report remains unresolved. The correction has static source review; corrected physical behavior still needs observation.

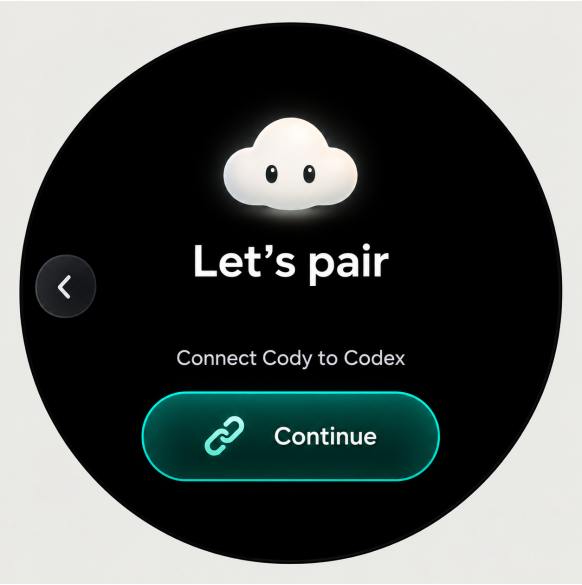
Preservation

All 13 preceding pages, prior errata and unrelated content are retained. This addendum supersedes only the cited escape/gesture behavior. Saved credentials are not erased.

Cody action language

IMPORTANT UI DECISION / approved 09 October 2026 / binding for future screens





Layout rule

Two actions: rounded cards side by side. Three: the same pair plus one centered action below. One: a wide centered capsule with its icon left of the label.

Shared visual language

Dark restrained gradients, thin semantic borders, white labels and action icons. Teal: affirmative/Codex; blue: network; graphite: No/Later/Options; amber: Retry/Cancel.

Circular geometry

Keep Back separate at the left. Share rendering and touch bounds in every orientation. QR pages keep the quiet zone and use the compact single-action variant.

Options and touch continuity

Setup keeps Network setup and Connect Codex distinct. Options opens screen 10.

41 / Setup

Left card: Connect Codex. Right card: Network setup. Centered third action: Options. Options opens 10 / Make it yours, then 36 and 11 / skin. Done returns to Setup; Back retraces this local preferences path. No network or Codex association is erased or implicitly started.

40 / Connect Codex? and 42 / No connection

Yes/No and Network setup/Later are side by side, with purpose-specific icons and colors. Their existing explicit backend admission stays intact. Screen IDs 05-15 and 36 share the action styling where an action is present.

Observed freeze / original cause

An authorized 90-second reset/capture recorded Yes at 16497 ms, entry to Connect Codex at 16498 ms, then a CST9217 report with fifteen FF bytes and ACK=FF at 17112 ms (expected AB). ESP_ERR_INVALID_RESPONSE stopped the renderer. LAN/IP was ready.

Scoped runtime correction

The touch owner still rejects malformed reports and publishes the real error. After startup only, that acquisition error is optional for rendering. Discard queued gesture reports, cancel the gesture and require a real acquired release before another press. No release is invented. The same acquisition error can mean an I2C NACK; its origin is not inferred. Startup, other acquisition errors, locks and route failures remain terminal.

Evidence boundary

Native source integration and approved mockups are documented here. The underlying cause of the FF report and corrected physical touch/standby behavior remain unverified. A successful build/flash alone does not establish those outcomes.

Errata and corrections

Revision date: 09 October 2026 / important UI decision and touch correction

Important design decision

The user approved the two-column pair plus centered third action, and the wide single action with icon to the left. These are binding layouts from this revision onward. Gradients, icons and semantic colors are required across Cody screens.

Affected original material

Original pages 01 (scope), 06 (retained link) and 07 (rediscovery); screens 12-16. Existing protocol/security qualifications remain in force. Also affected: screens 05-15, 35B, 36 and 40-42. Earlier stacked-blue control illustrations are superseded only in their action layout/styling.

Original analysis gaps

The initial analysis omitted Options re-entry and did not distinguish an optional malformed runtime touch report from renderer failure. This was a buco di analisi iniziale. Options now opens 10 and returns to Setup after preferences/skin.

Corrections and limits

Yes/No become horizontal and semantic. The measured invalid touch report retains its original error while rendering continues with interrupted gestures. Hardware FF root cause and physical corrected behavior remain open.

Preservation

All 15 preceding pages, earlier errata and unrelated content remain unchanged. Approved images are design references, not physical device screenshots.